Saumitra Sapre

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Education

University of California, San Diego Sep. 2022 – Jun. 2024
Master of Science, Computer Science and Engineering; GPA: 3.91/4.0 *San Diego, CA, USA*

Pune Institute of Computer Technology, Pune University Jul. 2017 – Jul. 2021
Bachelor of Engineering, Computer Engineering; GPA: 9.63/10 *Pune, MH, India*

Technical Skills

Languages: C/C++, Python, Rust, Java, GoLang, Shell & PowerShell scripting, SQL, R, HTML/CSS
Tools : Protocol Buffers, Docker, Kubernetes, gRPC, Vitis HLS, Git
Frameworks: NLTK, Pandas, Keras, TensorFlow, PyTorch,Scikit learn, NumPy, Rasa NLU, Android, ElectronJS
Skills: Storage Development, NLP, Android App Development, Game Development, High Level Synthesis
OS Modules: Linux Volume Management, ZFS, Windows Management Instrumentation, Microsoft VSS

Experience

Robin.io, A Rakuten Symphony Company Nov. 2026 – Present
Technical Lead *San Mateo, CA, USA*

- Engineered several fundamental features of Rakuten Cloud's S3-compatible Kubernetes-deployed object storage solution.

Software Engineer Nov. 2024 – Nov. 2026

- Designed and developed server-side encryption in the datapath, with a scalable key rotation subsystem integrated with commodity KMS providers.
- Built a flexible authentication system to facilitate external identity providers like LDAP/AD.
- Developed a zero-downtime disk and node expansion/evacuation architecture for seamless hardware refreshes.
- Completely redesigned user, policy, and access key management for a secure and streamlined access control experience.
- Implemented numerous stability enhancements and performance improvements.

Design Lab - UC San Diego Oct. 2022 – Jun. 2024
Graduate Researcher *San Diego, CA, USA*

- Analyzed AI explainability and trust in self-driving cars, contributing to key design decisions linking personal characteristics to vehicle informatics - reducing stress, improving design effectiveness, situational awareness, and driving outcomes.
- Used VR integration with **Unreal Engine 4** and **CARLA** to design **60+** realistic driving simulation scenarios.

RackWare Inc. Jul. 2021 – Aug. 2022
Software Engineer - RMM Team *Pune, MH, India*

- Part of the **5 person core engineering team** in India that developed, maintained and enhanced RackWare's premier cloud migration and DR product called 'RackWare Management Module' (RMM).
- Optimized RMM's storage safety mechanism to continuously monitor and securely cut off running syncs to prevent image corruption during high disk usage. Eliminated redundant R calls by **80%**, reducing major CPU load.
- Coordinated the complete overhaul of RMM's self-extracting installer, reducing redundant user interactions by **87%**. Designed several critical fail-safes to improve product robustness during installation.
- Set up a customized CLI automation suite using Python's Nose framework, specifically made for running complex sanity test suites by the engineering team.
- Added product support for auto-provisioning Ubuntu - 14,16,18, SUSE - 12, 15 images on AWS, and provisioning Oracle Linux hosts on Azure.
- Solved **20+** critical cloud migration issues across Azure, OCI and AWS environments.

Publications

Knowledge Graph Generation From Text Using Neural Machine Translation Techniques Aug. 2021
International Conference on Communication Information and Computing Technology (ICCICT) *Link*

- Analyzed several approaches used for SPARQL machine translation, particularly for data insertion in knowledge graphs. Curated custom datasets from DBpedia for machine translation of plain text to SPARQL insert queries which reached average **BLEU scores of 98.28** on Attention-RNN and Transformer models.

Projects

Surfstore - Fault Tolerant File Storage System | *gRPC, Protocol Buffers, GoLang*

- Implemented a fault tolerant file storage system (similar to Dropbox) to synchronise file access across multiple devices.
- Ensured fault tolerance and scalability of nodes by employing the **RAFT protocol** and **consistent hashing**.

Emotionally Aware Chatbot | *Rasa NLU, TensorFlow, MongoDB, Python* | *Link*

- **Led a team of 4** that designed a user adaptive conversational agent to perform cognitive behavioral therapy.
- Modified the DIET classifier and implemented a custom TensorFlow NMT engine to detect intent and entities and curate personalized, empathetic responses that helped improve mood.
- Presented project at the International Conference on Smart Technology, AI & Computer Engineering (**ICSTAICE**)

ML Performance Analysis on Different Hardware | *Keras, HLS4ML, AMD Vitis HLS, Python*

- Analysed the performance of standard ML models across Intel Xeon (CPU), Nvidia Tesla (GPU) and Zynq-7000 SoC (FPGA) architectures. Achieved a **600x** speed improvement over the CPU and **350x** speed improvement over the GPU when testing the same models on the FPGA.
- Performed model compression and layer quantization optimizations to fit the model within FPGA resource constraints.

Indoor Positioning System | *Android, BLE, Java* | *Link*

- Built a system to navigate between places in closed environments using Bluetooth Low Energy (BLE) Technology.
- Performed real-time localization using **trilateration** & route mapping with the help of BLE beacons placed in closed environments where GPS was ineffective. Accurately mapped **100 sq. ft.** of college library interiors.
- Ranked amongst **top 5 winners** in the institute level round at Smart India Hackathon (SIH) 2020.